

Polynomial Regression Assignment

Dayal Gupta
BT2024167
GitHub Repository

1 Introduction

This assignment uses polynomial regression to predict a continuous target from two personalized datasets. VAR1 concerns power plant steam turbine optimization, and VAR2 concerns thermal reservoir mapping. In both datasets, the target column is called y .

The test files do not contain target values. I used the training data for model selection and used the test features only to produce the final predictions.

2 Data

VAR1 has six inputs, $x_1, \dots, x_6$, and VAR2 has three inputs, x_1, x_2, x_3 . Each variant has 1,000 training rows and 1,000 test rows. The training files contain the target y ; the test files contain only the input columns. There were no missing values, and the train and test feature columns matched for each variant.

For VAR1, the inputs describe turbine operating parameters and y is the Net Power Score. For VAR2, the inputs describe east-west position, north-south position, and vertical depth; y is the Thermal Anomaly Score.

3 Method

I expanded the input features with polynomial terms, including interactions, and fitted regression models to the expanded data. The search compared ordinary least squares (OLS), Ridge, Lasso, and ElasticNet models. For regularized models, I also compared runs with and without standard scaling.

The degree ranges were 2–7 for VAR1 and 6–14 for VAR2. Model selection used nested cross-validation on the training data: five outer folds estimated performance, while three inner folds selected regularization settings. The folds were shuffled with random state 7. Mean squared error (MSE) was the main selection metric; R^2 was also recorded.

I used a one-standard-error rule when choosing among close results. This favours a simpler model when its cross-validation score is within one standard error of the lowest MSE. The test data was not used to choose the degree or tune model settings.

4 Results and Model Selection

Table 1 gives the selected model for each variant and its outer-fold cross-validation results. The OLS values are comparison baselines using degree 4 for VAR1 and degree 8 for VAR2.

Table 1: Selected models and OLS baselines from outer-fold cross-validation.

Variant	Selected model	Degree	Alpha	Selected CV MSE	OLS baseline MSE
VAR1	Scaled Lasso	5	0.006	0.3491 ± 0.0384	0.7717
VAR2	Ridge	10	0.1	0.2329 ± 0.0134	0.2556

The selected VAR1 model is a degree-5 Lasso model with standard scaling. Its mean outer-fold R^2 was 0.9631. The final fit retained 150 nonzero polynomial coefficients. Its cross-validation MSE was lower than the degree-4 OLS baseline in this experiment.

The selected VAR2 model is a degree-10 Ridge model without scaling. Its mean outer-fold R^2 was 0.9943. Degree 11 Ridge had a slightly lower mean CV MSE (0.2327), but degree 10 was within one standard error of that result and has a smaller polynomial basis. I therefore chose degree 10 under the one-standard-error rule. Its CV MSE was also lower than the degree-8 OLS baseline.

These values are estimates from folds of the training data. They do not give the MSE or R^2 on the hidden test targets.

5 VAR1 Figures

The degree comparison shows the cross-validation error across the tested polynomial degrees. The other plots show out-of-fold predictions and residuals for the selected model.

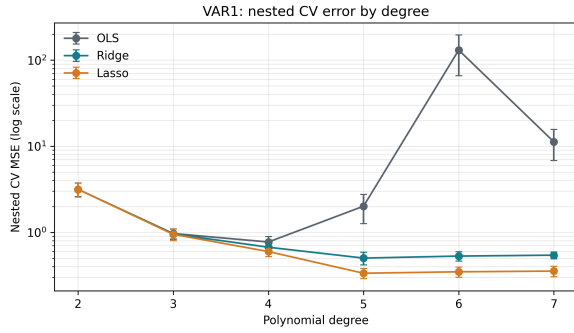


Figure 1: VAR1 cross-validation MSE by polynomial degree.

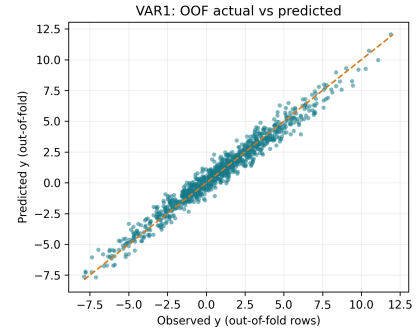


Figure 2: VAR1 out-of-fold actual values and predictions.

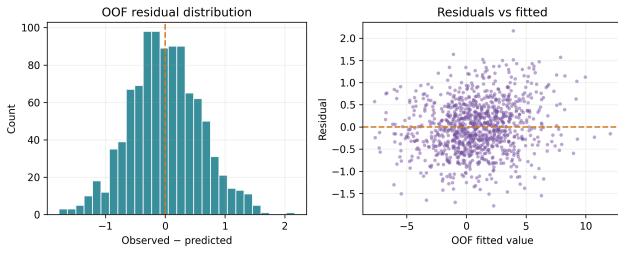


Figure 3: VAR1 residual diagnostics for the selected model.

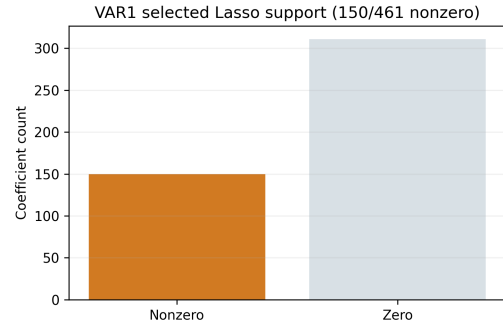


Figure 4: Nonzero coefficients retained by the VAR1 Lasso model.

6 VAR2 Figures and Stability Check

The VAR2 degree comparison and out-of-fold plots summarize the selected model's training-only evaluation. I also checked high-degree OLS on a separate 80/20 split of the VAR2 training data, using random state 42.

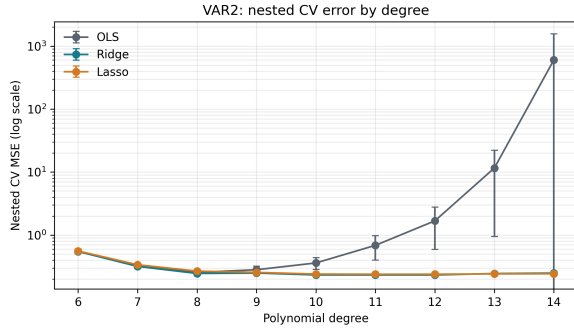


Figure 5: VAR2 cross-validation MSE by polynomial degree.

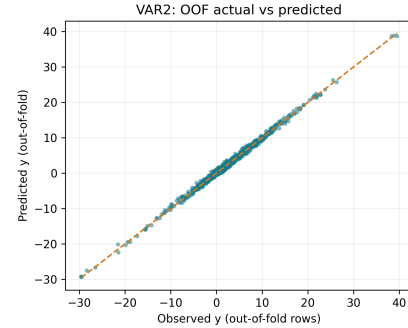


Figure 6: VAR2 out-of-fold actual values compared with predictions.

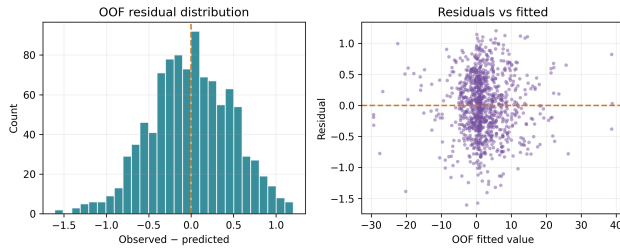


Figure 7: VAR2 residual diagnostics for the selected model.

At degree 15, the OLS design matrix had rank 793 out of 815 columns, and its smallest-to-largest singular-value ratio was 6.8×10^{-17} . At degree 20, it had rank 793 out of 1,770 columns; the ratio was 7.6×10^{-17} , and the validation MSE was about 29.9 million. This check showed numerical instability in high-degree OLS on this split.

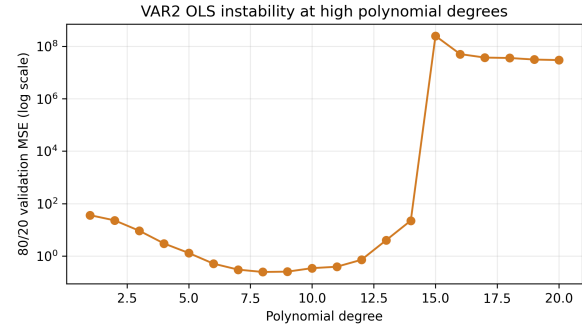


Figure 8: VAR2 OLS validation MSE by polynomial degree.

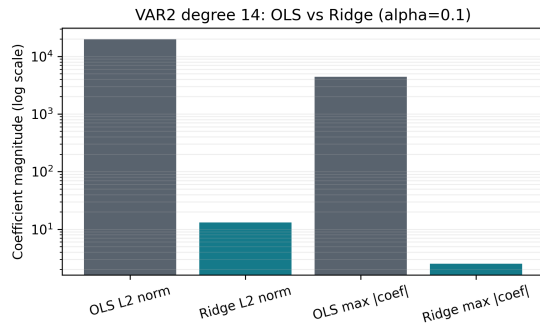


Figure 9: Coefficient sizes for degree-14 VAR2 OLS and Ridge models.

7 Final Models and Predictions

After selecting the configurations, I refit each model using all 1,000 training rows for that variant. I then predicted y for all 1,000 rows in the corresponding test file, keeping the original row order.

The final configurations were:

- VAR1: degree-5 polynomial features, standard scaling, and Lasso with $\alpha = 0.006$.
- VAR2: degree-10 polynomial features and Ridge with $\alpha = 0.1$, without scaling.

The predictions were saved as `BT2024167_pred_var1.csv` and `BT2024167_pred_var2.csv`. Each file contains one column, y , and no saved index column.

8 Conclusion

I selected scaled degree-5 Lasso for VAR1 and degree-10 Ridge for VAR2 based on the training-only nested cross-validation results. The VAR2 one-standard-error rule favoured degree 10 over degree 11, which had a slightly lower mean CV MSE. Both final models were refit on all available training rows before making test predictions. Since test labels are hidden, their final performance cannot be reported here.